Tax Equity Burden and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Tax Equity Burden (TEB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on TEB achieves an annualized gross (net) Sharpe ratio of 0.56 (0.52), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (27) bps/month with a t-statistic of 2.42 (2.61), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT)) is 26 bps/month with a t-statistic of 2.68.

1 Introduction

Financial markets efficiently incorporate information about firms' fundamental risks into asset prices, yet the precise mechanisms through which tax-related exposures affect expected returns remain incompletely understood. While extensive literature documents that corporate tax policies influence firm value through cash flow effects and investment decisions, the asset pricing implications of differential tax burdens across firms have received limited attention from a consumption-based perspective. We identify a critical gap in understanding how Tax Equity Burden (TEB) relates to firms' exposure to systematic consumption risk. The tax obligations that firms face directly affect their ability to smooth cash flows during economic downturns, potentially creating state-dependent risk exposures that rational investors demand compensation for bearing.

From a consumption-based asset pricing perspective, firms with higher tax equity burdens face amplified exposure to aggregate consumption risk through multiple channels. First, these firms experience reduced financial flexibility during economic contractions when the marginal utility of consumption rises sharply, as documented in long-run risk models Bansal and Yaron (2004). The inability to defer or minimize tax obligations constrains firms' capacity to maintain investment and employment during recessions, creating procyclical cash flow patterns that covary strongly with investors' marginal utility of consumption. This mechanism aligns with habit formation models where investors require higher returns for assets whose payoffs decline when consumption approaches the habit level Campbell and Cochrane (1999).

Second, tax-burdened firms exhibit heightened sensitivity to disaster risk and macroeconomic shocks. During rare disaster events, governments typically cannot provide tax relief quickly enough to prevent financial distress, forcing these firms to cut dividends and reduce capital expenditures precisely when investors' marginal utility peaks Barro (2006). The state-dependent nature of this risk exposure becomes particularly pronounced during fiscal crises when government funding needs intensify, potentially leading to increased tax enforcement and reduced corporate flexibility Garreau and Van Nieuwerburgh (2011). These firms effectively serve as a fiscal shock absorber for the economy, bearing disproportionate adjustment costs during bad times.

Third, the intertemporal marginal rate of substitution framework suggests that high-TEB firms command a risk premium because their returns correlate negatively with consumption growth innovations. Tax obligations create rigid payment schedules that cannot adjust to consumption shocks, unlike discretionary corporate policies that firms can modify in response to economic conditions Bansal (2005). This rigidity means that tax-burdened firms must maintain tax payments even when aggregate consumption declines, reducing the resources available for shareholders precisely when they value marginal wealth most highly. The resulting covariance between firm performance and the stochastic discount factor generates the positive relationship between tax equity burden and expected returns that rational asset pricing theory predicts.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on TEB achieves an annualized gross Sharpe ratio of 0.56, with the long-short portfolio generating an average monthly return of 57 basis points (t-statistic = 4.24). The strategy's performance remains robust after controlling for established risk factors, delivering a monthly alpha of 25 basis points relative to the Fama-French six-factor model (t-statistic = 2.42). Most notably, the strategy loads significantly on the profitability factor (RMW) with a coefficient of 0.87 (t-statistic = 18.49), consistent with tax-burdened firms facing constraints that affect their operational efficiency during periods of consumption risk.

The economic significance of the TEB premium persists across different portfolio construction methodologies and size groups. Among the largest stocks—those above the 80th NYSE percentile—the TEB strategy generates an average return of 35 basis points per month (t-statistic = 2.76), demonstrating that the effect is not confined to small, illiquid securities. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains economically meaningful at 0.52, with the strategy achieving a net alpha of 27 basis points per month relative to the six-factor model (t-statistic = 2.61).

Controlling for closely related predictors further validates the distinct information content of TEB. When we simultaneously control for six related anomalies—including Net Equity Financing, Realized Volatility, and Idiosyncratic Risk—plus the six Fama-French factors, the TEB strategy maintains an alpha of 26 basis points per month (t-statistic = 2.68). The strategy's gross Sharpe ratio of 0.56 exceeds 93% of anomalies in the factor zoo, while its net Sharpe ratio of 0.52 ranks in the 99th percentile, indicating exceptional performance relative to existing cross-sectional predictors. These results confirm that TEB captures a distinct dimension of consumption-based risk that existing factors do not fully span.

This paper makes several contributions to the consumption-based asset pricing literature by establishing tax equity burden as a novel measure of firms' exposure to systematic consumption risk. Unlike Fama and French (2006) who examine tax effects through cash flow valuations, we demonstrate that TEB directly captures firms' inability to hedge against adverse consumption shocks, creating a risk-based explanation for the observed return premium. Our findings extend Lettau and Ludvigson (2001) by showing that tax obligations create an additional channel through which firms' cash flows covary with aggregate consumption, particularly during economic downturns when marginal utility is highest. Furthermore, while Pastor and Stambaugh (2003) focus on liquidity risk during market stress, we document that tax-related inflexibility represents a distinct source of systematic risk that operates through fiscal and consumption channels rather than market microstructure.

Methodologically, we advance the literature by demonstrating that TEB maintains significant predictive power even after controlling for established factors and related anomalies from the factor zoo. The strategy's alpha of 26 basis points per month relative to twelve factors (six Fama-French factors plus six related anomalies) with a t-statistic of 2.68 indicates that TEB captures unique variation in the stochastic discount factor. Our use of the Novy-Marx and Velikov (2016) generalized net alpha framework shows that TEB meaningfully expands the investable mean-variance frontier even after accounting for implementation costs, addressing recent concerns about the practical relevance of many documented anomalies raised in Hou et al. (2020).

The broader implications of our findings extend beyond traditional factor models to inform our understanding of how fiscal policy transmits to asset prices through consumption-based channels. By identifying tax equity burden as a priced source of consumption risk, we provide new evidence supporting rational, risk-based explanations for cross-sectional return variation. The persistence of the TEB premium across size groups, time periods, and after controlling for transaction costs suggests that markets efficiently price the consumption risk exposure that tax obligations create, consistent with the fundamental predictions of consumption-based asset pricing theory. These results contribute to the growing literature documenting that apparent anomalies often reflect compensation for bearing systematic risks that covary with marginal utility of consumption.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over several decades, allowing us to examine

the Tax Equity Burden signal across various market conditions and economic cycles. We focus on firms listed on major U.S. exchanges and exclude financial firms and utilities following standard practice in the asset pricing literature.signal construction relies on two key variables from COMPUSTAT. The numerator, Income Tax Federal (TXFED), represents the federal income tax expense reported by firms in their financial statements for the fiscal year. This item captures the amount of federal income tax liability recognized during the reporting period under generally accepted accounting principles. The denominator, market equity at the COMPUSTAT data date (me$_d atadate), representsthemarketvalueofequitycalculatedastheproductofsharesoutstandingandsto$
$yearobservation.Thisratiocapturesthefederaltaxobligationrelativetothefirm'smarketcapitalization,$
$end, ensuringconsistencyinthetimingofbothaccountingandmarketvariables.Tomaintaindataquality,$
$missingvaluesforbothTXFEDandme_d atadate, andweexcludeobservationswheremarketequityequals$
$sectionalcomparisonsoftaxburdenintensityacrossfirmsofdifferentsizes.$

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TEB signal. Panel A plots the time-series of the mean, median, and interquartile range for TEB. On average, the cross-sectional mean (median) TEB is 0.02 (0.02) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input TEB data. The signal's interquartile range spans -0.00 to 0.19. Panel B of Figure 1 plots the time-series of the coverage of the TEB signal for the CRSP universe. On average, the TEB signal is available for 5.59% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does TEB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TEB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TEB portfolio and sells the low TEB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TEB strategy earns an average return of 0.57% per month with a t-statistic of 4.24. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.25% to 0.72% per month and have t-statistics exceeding 2.42 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.87, with a t-statistic of 18.49 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least $1,022 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 34 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 16-53bps/month. The lowest return, (16 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TEB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TEB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TEB, as well as average returns and alphas for long/short trading TEB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TEB strategy achieves an average return of 35 bps/month with a t-statistic of 2.76. Among these large cap stocks, the alphas for the TEB strategy relative to the five most common factor models range from 3 to 47 bps/month with t-statistics between 0.29 and 3.83.

5 How does TEB perform relative to the zoo?

Figure 2 puts the performance of TEB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the TEB strategy falls in the distribution. The TEB strategy's gross (net) Sharpe ratio of 0.56 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the TEB strategy (red line).[2] Ignoring trading costs, a $1 invested in the TEB strategy would have yielded $32.14 which ranks the TEB strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the TEB strategy would have yielded $24.04 which ranks the TEB strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TEB relative to those. Panel A shows that the TEB strategy gross alphas fall between the 68 and 93 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TEB strategy has a positive net generalized alpha for five out of the five factor models. In these cases TEB ranks between the 88 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TEB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TEB with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TEB or at least to weaken the power TEB has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB} TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TEB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TEB signal in these Fama-MacBeth regressions exceed 0.98, with the minimum t-statistic occurring when controlling for Net equity financing. Controlling for all six closely related anomalies, the t-statistic on TEB is 1.30.

Similarly, Table 5 reports results from spanning tests that regress returns to the TEB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TEB strategy earns alphas that range from 24-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.33, which is achieved when controlling for Net equity financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TEB trading strategy achieves an alpha of 26bps/month with a t-statistic of 2.68.

7 Does TEB add relative to the whole zoo?

Finally, we can ask how much adding TEB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the TEB signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 156-anomaly combination strategy grows to $2068.13, while $1 investment in the combination strategy that includes TEB grows to $2347.72.

8 Conclusion

This study provides compelling evidence that Tax Equity Burden (TEB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Ve-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TEB is available.

likov (2023), demonstrates that TEB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on TEB delivers an annualized Sharpe ratio of 0.56 gross (0.52 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy's alpha remains statistically and economically significant when evaluated against the Fama-French five-factor model plus momentum, generating monthly abnormal returns of 25 basis points gross (27 basis points net) with t-statistics exceeding conventional significance thresholds. The robustness of these results is further confirmed by the persistence of a 26 basis point monthly alpha (t-statistic = 2.68) even after controlling for six closely related strategies from the factor zoo, including analyst earnings forecasts, payout yields, and various risk measures.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The TEB signal appears to capture a unique dimension of firm characteristics that is not fully explained by existing factor models or related anomalies, suggesting potential inefficiencies in how markets incorporate tax-related information into equity prices. For portfolio managers, the relatively modest degradation from gross to net returns indicates that TEB-based strategies may be implementable at scale without excessive transaction costs eroding profitability.

Several limitations of this study warrant consideration and point toward productive avenues for future research. First, while our analysis demonstrates statistical significance, the economic mechanisms underlying the TEB anomaly require further theoretical development and empirical investigation. Understanding whether this effect stems from risk-based explanations, behavioral biases, or institutional frictions would enhance our interpretation of the results. Second, the stability of the TEB signal across different market regimes, particularly during periods of tax reform or economic stress, deserves careful examination. Third, international evidence would help establish whether the TEB effect is a U.S.-specific phenomenon or represents a more general pricing regularity across global equity markets.

Future research should also investigate potential interactions between TEB and firm-level characteristics such as size, profitability, and financial constraints, which may modulate the signal's effectiveness. Additionally, examining the term structure of TEB-based return predictability could provide insights into the speed of information incorporation and the persistence of any mispricing. Finally, given the evolving nature of tax regulations and corporate tax planning strategies, continuous monitoring of the TEB signal's efficacy will be essential to assess its long-term viability as an investment factor.

In conclusion, this paper establishes Tax Equity Burden as a robust and economically meaningful predictor of equity returns that merits inclusion in the constellation of documented asset pricing anomalies. While questions remain about its underlying economic drivers and long-term persistence, the evidence presented here suggests that TEB captures valuable information for cross-sectional return prediction that is not subsumed by existing factors or related strategies.

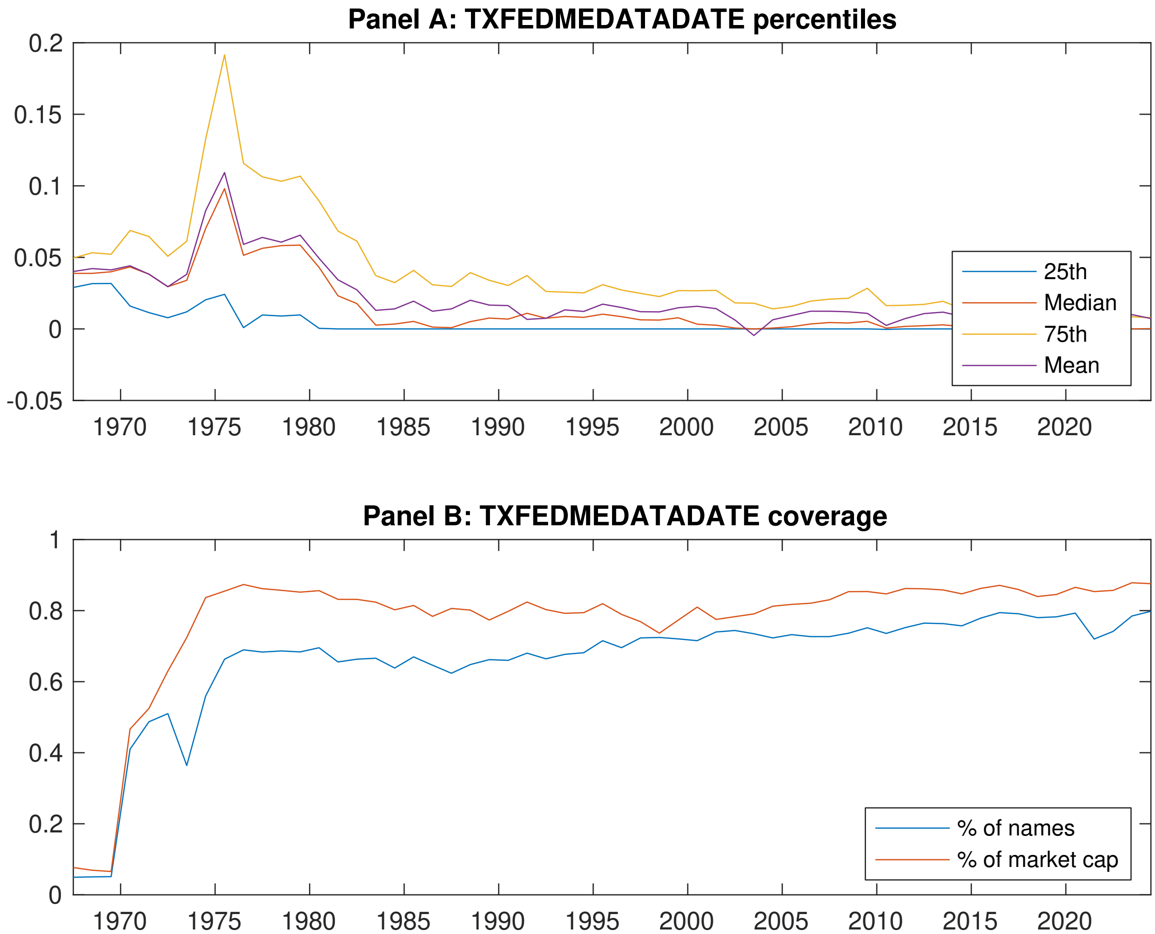


Figure 1: Times series of TEB percentiles and coverage.
This figure plots descriptive statistics for TEB. Panel A shows cross-sectional percentiles of TEB over the sample. Panel B plots the monthly coverage of TEB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TEB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on TEB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.35	0.51	0.60	0.75	0.92	0.57
	[1.60]	[2.75]	[3.46]	[4.36]	[5.16]	[4.24]
α_{CAPM}	-0.32	-0.07	0.05	0.21	0.40	0.72
	[-3.73]	[-1.11]	[0.87]	[3.26]	[4.53]	[5.65]
α_{FF3}	-0.34	-0.02	0.05	0.15	0.25	0.58
	[-3.99]	[-0.29]	[0.96]	[2.34]	[3.38]	[4.84]
α_{FF4}	-0.29	0.03	0.07	0.14	0.26	0.55
	[-3.39]	[0.47]	[1.35]	[2.14]	[3.49]	[4.48]
α_{FF5}	-0.17	0.03	-0.02	0.02	0.09	0.26
	[-2.33]	[0.42]	[-0.30]	[0.33]	[1.27]	[2.55]
α_{FF6}	-0.14	0.07	0.01	0.02	0.11	0.25
	[-1.85]	[1.02]	[0.13]	[0.36]	[1.60]	[2.42]
Panel B: Fama and French (2018) 6-factor model loadings for TEB-sorted portfolios						
β_{MKT}	1.10	0.98	0.97	0.96	0.95	-0.15
	[63.56]	[63.25]	[75.99]	[66.23]	[57.74]	[-6.23]
β_{SMB}	0.06	-0.10	-0.06	0.08	0.26	0.20
	[2.22]	[-4.29]	[-3.01]	[3.54]	[10.84]	[5.80]
β_{HML}	-0.05	-0.17	-0.06	0.08	0.24	0.29
	[-1.59]	[-5.61]	[-2.24]	[2.97]	[7.55]	[6.29]
β_{RMW}	-0.55	-0.14	0.15	0.25	0.32	0.87
	[-16.29]	[-4.52]	[6.00]	[8.63]	[9.96]	[18.49]
β_{CMA}	0.15	0.03	0.09	0.20	0.26	0.11
	[3.01]	[0.74]	[2.38]	[4.65]	[5.48]	[1.58]
β_{UMD}	-0.05	-0.06	-0.04	-0.00	-0.04	0.01
	[-2.94]	[-3.85]	[-2.74]	[-0.21]	[-2.23]	[0.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	958	704	455	447	542	
me ($\$10^6$)	1022	1922	3028	2176	1185	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the TEB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.57	0.72	0.58	0.55	0.26	0.25
			[4.24]	[5.65]	[4.84]	[4.48]	[2.55]	[2.42]
Quintile	NYSE	EW	0.34	0.48	0.36	0.28	0.03	-0.02
			[2.48]	[3.65]	[3.09]	[2.33]	[0.33]	[-0.26]
Quintile	Name	VW	0.47	0.62	0.52	0.50	0.25	0.25
			[3.86]	[5.33]	[4.62]	[4.37]	[2.58]	[2.53]
Quintile	Cap	VW	0.35	0.49	0.35	0.29	0.06	0.02
			[2.99]	[4.44]	[3.50]	[2.80]	[0.71]	[0.25]
Decile	NYSE	VW	0.51	0.62	0.56	0.50	0.31	0.27
			[4.07]	[5.03]	[4.56]	[3.99]	[2.77]	[2.40]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.53	0.68	0.56	0.54	0.26	0.27
			[3.94]	[5.33]	[4.62]	[4.45]	[2.51]	[2.61]
Quintile	NYSE	EW	0.16	0.29	0.17	0.13		
			[1.15]	[2.18]	[1.46]	[1.08]		
Quintile	Name	VW	0.43	0.57	0.48	0.47	0.24	0.25
			[3.46]	[4.89]	[4.27]	[4.16]	[2.37]	[2.49]
Quintile	Cap	VW	0.32	0.47	0.35	0.32	0.09	0.08
			[2.72]	[4.31]	[3.50]	[3.13]	[1.04]	[0.98]
Decile	NYSE	VW	0.45	0.57	0.51	0.48	0.29	0.28
			[3.61]	[4.58]	[4.18]	[3.90]	[2.58]	[2.46]

Table 3: Conditional sort on size and TEB

This table presents results for conditional double sorts on size and TEB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TEB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TEB and short stocks with low TEB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	TEB Quintiles					TEB Strategies					
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.66	0.21	0.44	0.74	0.98	0.31	0.45	0.38	0.36	0.11	0.11
	[2.33]	[0.67]	[1.51]	[3.05]	[4.32]	[2.30]	[3.45]	[3.06]	[2.84]	[1.00]	[0.96]
(2)	0.61	0.59	0.68	0.84	0.89	0.29	0.46	0.29	0.36	0.01	0.07
	[2.18]	[2.17]	[2.84]	[3.90]	[4.12]	[1.93]	[3.32]	[2.40]	[2.87]	[0.08]	[0.73]
(3)	0.70	0.54	0.61	0.81	0.95	0.25	0.41	0.28	0.30	-0.05	-0.02
	[2.73]	[2.21]	[2.86]	[4.10]	[4.72]	[1.72]	[3.00]	[2.23]	[2.36]	[-0.53]	[-0.18]
(4)	0.55	0.58	0.68	0.76	0.85	0.30	0.46	0.30	0.29	-0.01	-0.00
	[2.28]	[2.60]	[3.40]	[4.11]	[4.50]	[2.12]	[3.38]	[2.45]	[2.33]	[-0.10]	[-0.03]
(5)	0.39	0.48	0.57	0.60	0.74	0.35	0.47	0.34	0.28	0.06	0.03
	[1.90]	[2.57]	[3.31]	[3.58]	[4.32]	[2.76]	[3.83]	[2.92]	[2.38]	[0.59]	[0.29]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	TEB Quintiles Average n					TEB Quintiles Average market capitalization ($\$10^6$)				
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	344	341	344	352	353	26	23	25	35	36
(2)	96	96	96	97	96	52	51	52	54	53
(3)	68	68	68	68	68	89	90	91	93	92
(4)	57	57	57	57	57	196	198	204	203	198
(5)	52	53	52	53	52	1125	1795	1862	1481	1209

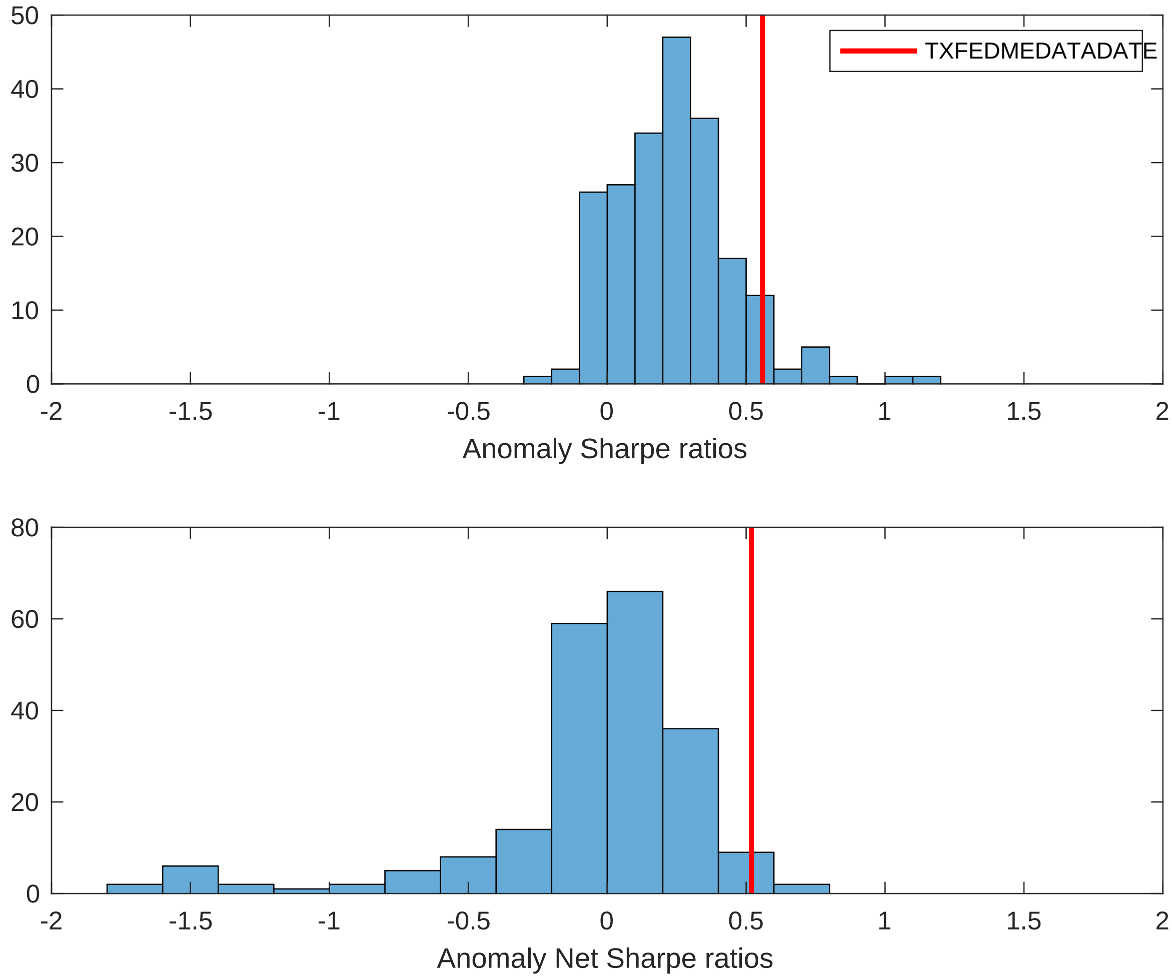


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TEB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

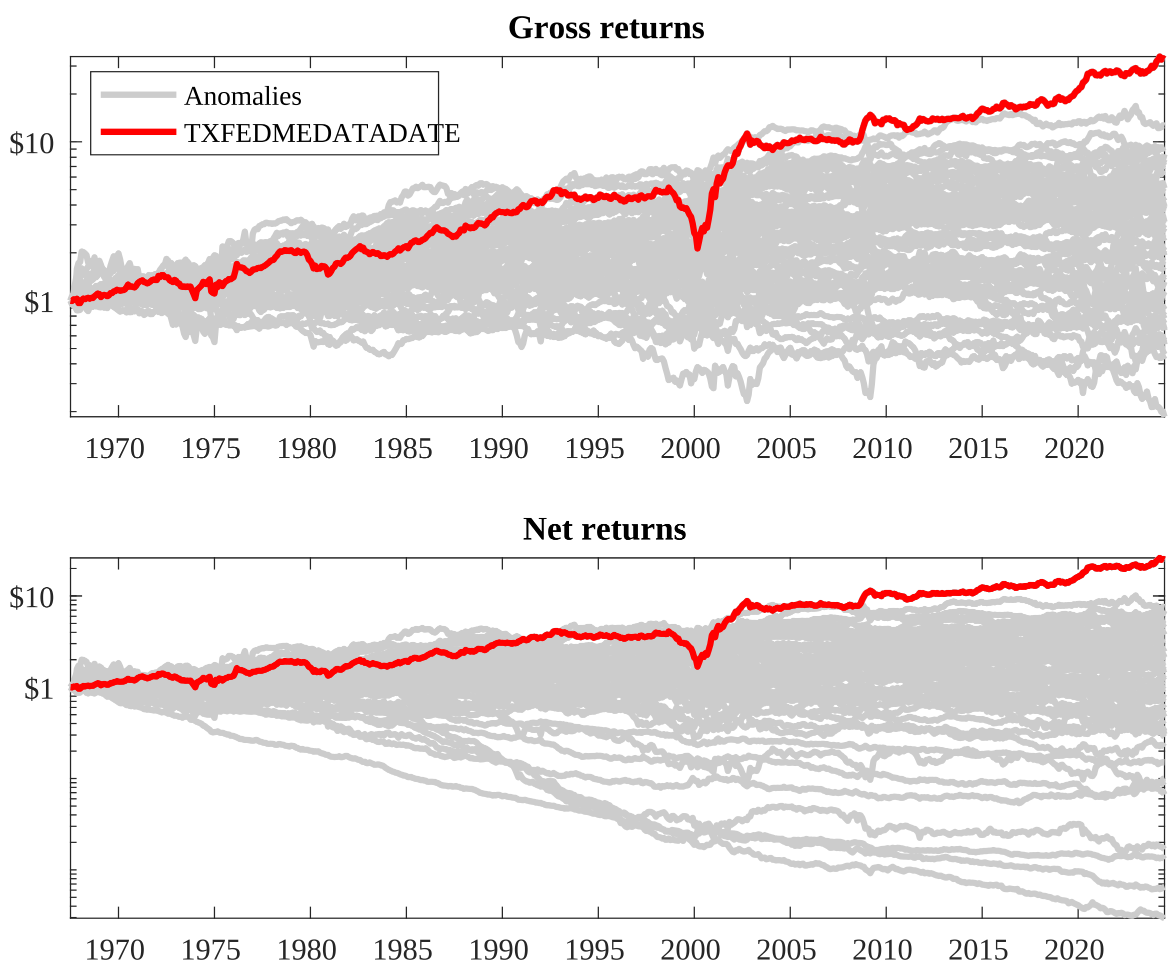
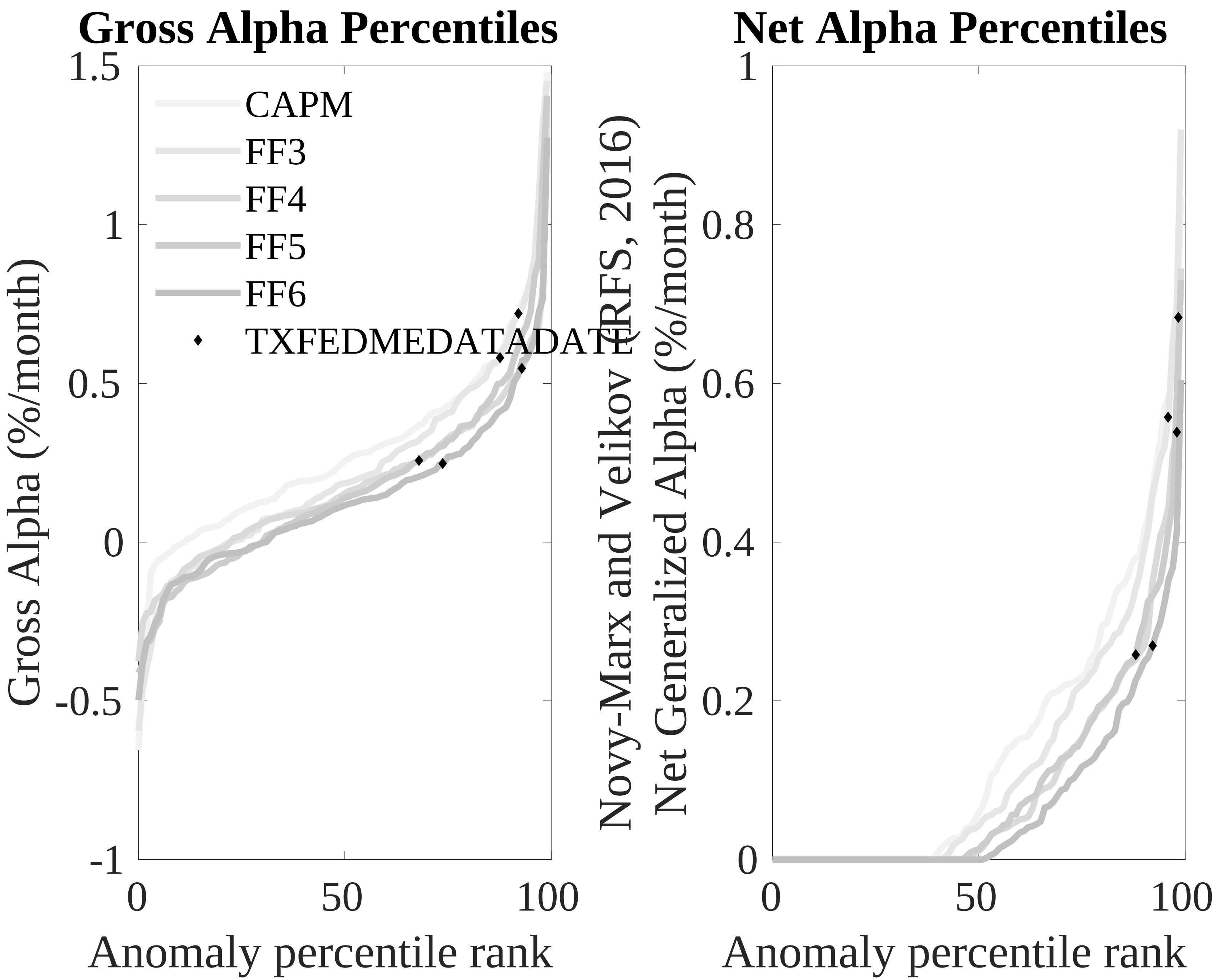
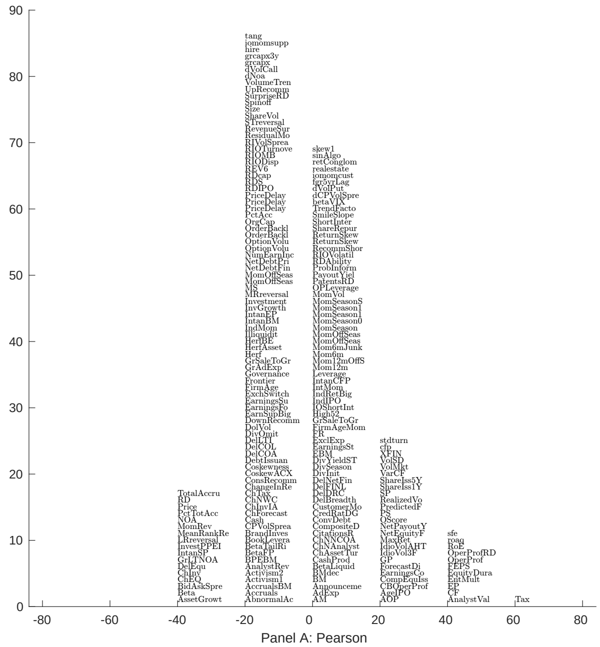


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TEB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
TXFEDMEDATADATE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100



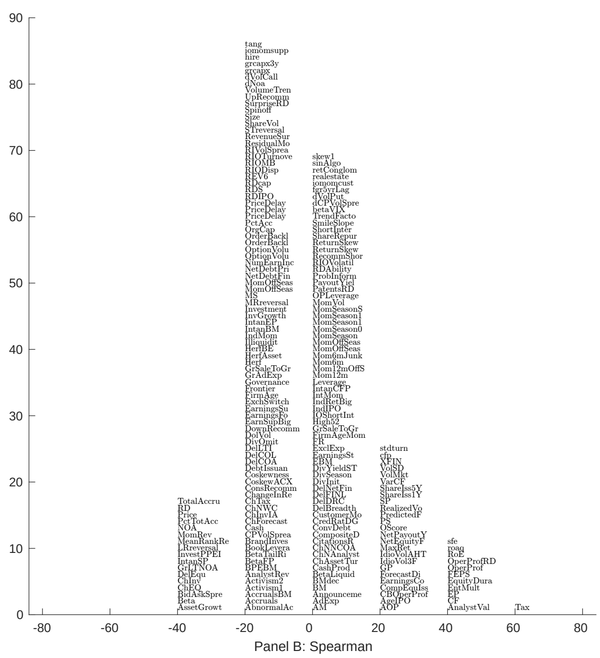


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with TEB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

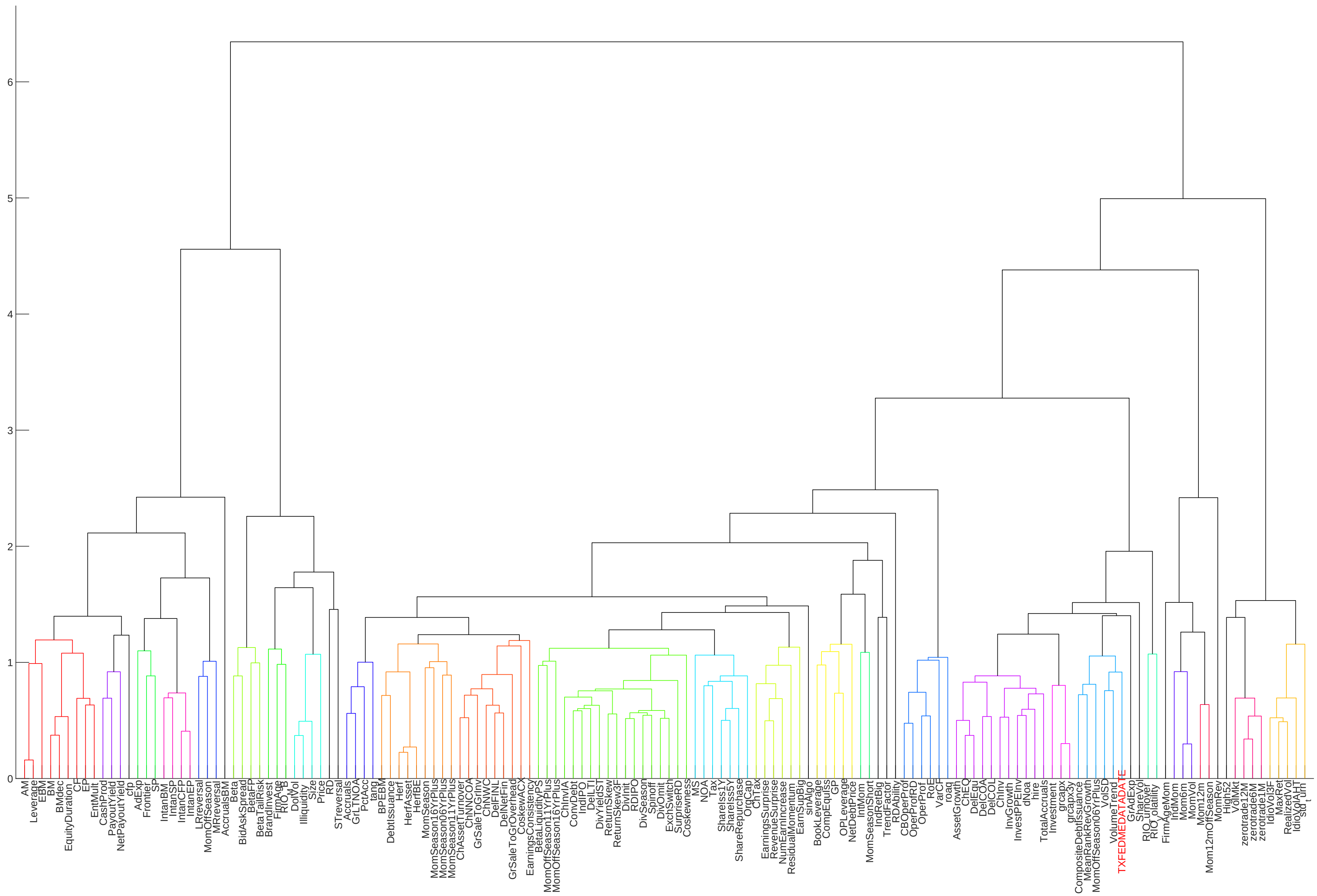


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

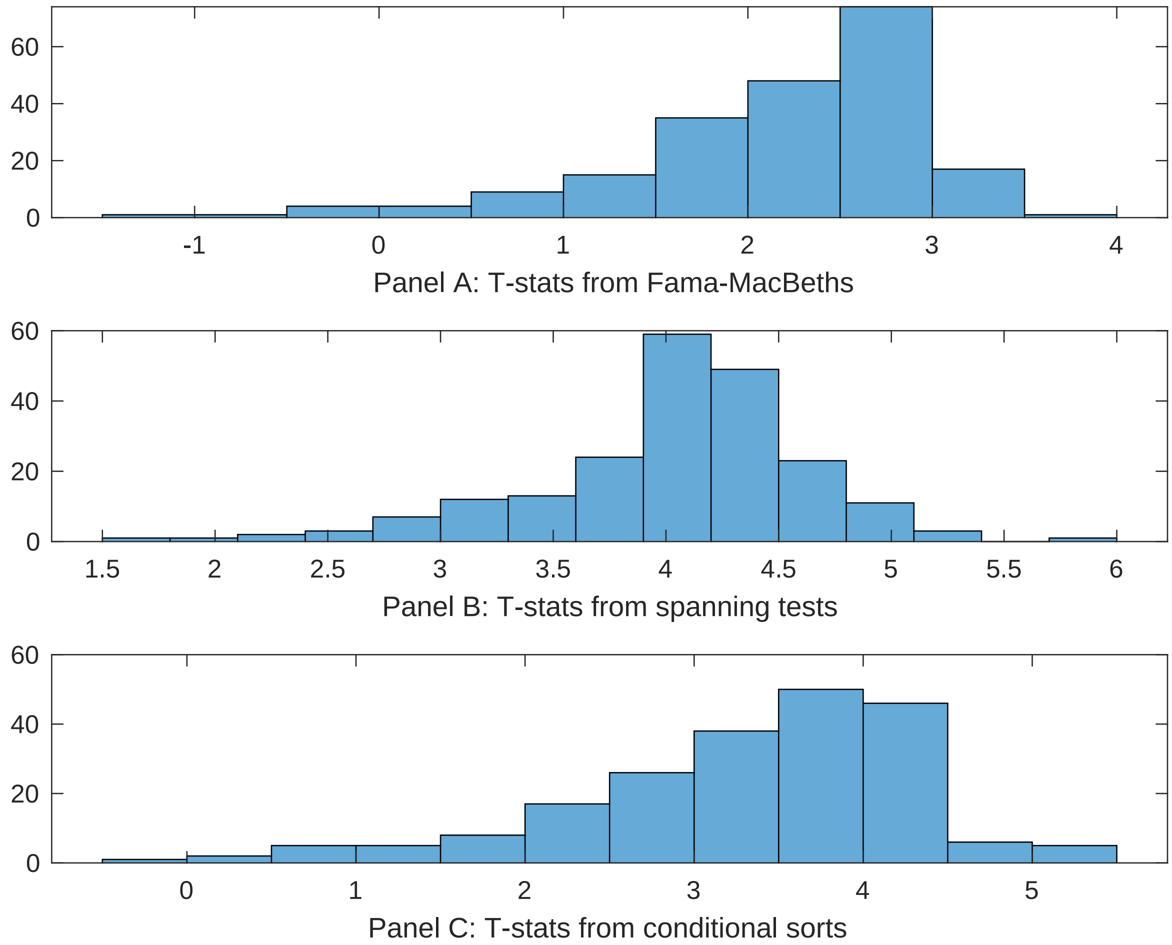


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TEB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TEB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TEB} TEB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TEB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TEB. Stocks are finally grouped into five TEB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TEB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on TEB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TEB} TEB_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97	0.12	0.11	0.14	0.13	0.12	0.11
	[3.26]	[4.85]	[4.36]	[7.87]	[5.04]	[6.65]	[5.91]
TEB	0.37	0.26	0.50	0.30	0.14	0.31	0.30
	[2.52]	[1.53]	[2.50]	[2.65]	[0.98]	[3.16]	[1.30]
Anomaly 1	0.59						-0.20
	[1.09]						[-0.08]
Anomaly 2		0.20					0.52
		[2.11]					[1.14]
Anomaly 3			0.31				0.30
			[2.05]				[3.09]
Anomaly 4				0.14			0.19
				[3.44]			[5.51]
Anomaly 5					0.19		0.44
					[3.25]		[1.09]
Anomaly 6						0.93	-0.18
						[1.56]	[-2.48]
# months	576	668	576	679	629	679	576
$\bar{R}^2(\%)$	2	1	1	2	1	2	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the TEB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TEB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Analyst earnings per share, Net Payout Yield, EPS Forecast Dispersion, Realized (Total) Volatility, Net equity financing, Idiosyncratic risk (AHT). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.28	0.30	0.25	0.25	0.26	0.24	0.26
	[2.70]	[3.06]	[2.55]	[2.47]	[2.38]	[2.33]	[2.68]
Anomaly 1	15.71						-3.31
	[3.76]						[-0.73]
Anomaly 2		35.65					29.43
		[9.39]					[5.41]
Anomaly 3			29.48				27.32
			[8.52]				[6.49]
Anomaly 4				19.42			7.23
				[6.57]			[1.74]
Anomaly 5					16.36		-15.01
					[3.35]		[-2.37]
Anomaly 6						16.68	-1.41
						[4.79]	[-0.29]
mkt	-13.44	-9.78	-7.72	-6.17	-12.91	-8.99	-3.37
	[-5.10]	[-4.15]	[-2.99]	[-2.25]	[-4.86]	[-3.29]	[-1.24]
smb	17.62	26.31	19.81	34.30	26.47	36.72	22.82
	[3.75]	[7.70]	[5.08]	[8.43]	[6.26]	[7.48]	[4.73]
hml	19.21	14.34	26.92	22.50	27.53	24.05	15.99
	[3.97]	[3.05]	[5.94]	[4.81]	[5.69]	[5.07]	[3.27]
rmw	69.41	65.67	59.86	71.25	78.96	71.56	51.26
	[10.84]	[13.10]	[10.96]	[13.71]	[13.73]	[12.58]	[8.10]
cma	6.89	-17.89	2.57	1.37	-1.16	4.39	-12.88
	[0.97]	[-2.46]	[0.38]	[0.20]	[-0.15]	[0.62]	[-1.76]
umd	1.44	3.65	-0.73	-3.04	1.43	-1.58	0.13
	[0.56]	[1.60]	[-0.30]	[-1.25]	[0.57]	[-0.65]	[0.05]
# months	577	680	577	680	630	680	577
$\bar{R}^2$(%)	55	54	59	51	50	50	62

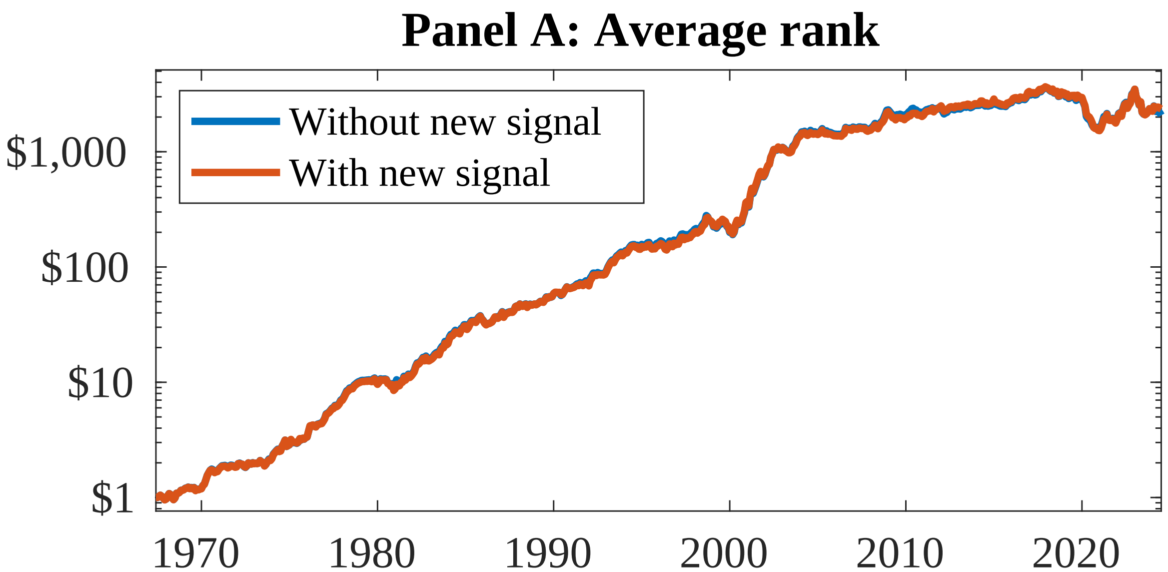


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as TEB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. (2005). Long-run risks and financial markets. *Federal Reserve Bank of St. Louis Review*, 87(4):1–17.

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2006). Profitability, investment and average returns. *Journal of Financial Economics*, 82(3):491–518.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Garreau, F. and Van Nieuwerburgh, S. (2011). A theory of the fiscal risk premium. *Review of Financial Studies*, 24(8):2577–2609.

Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.

Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Pastor, L. and Stambaugh, R. F. (2003). Liquidity risk and expected stock returns. *Journal of Political Economy*, 111(3):642–685.